Rotations

1. Euler Angle
    1. Euler Angle is very intuitive
    2. Euler Angle to Rotation matrix
        1. Rotation about principal axis is represented as:
        2. $R_x(\alpha) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\alpha & -\sin\alpha \\ 0 & \sin\alpha & \cos\alpha \end{bmatrix}$
        3. $R_y(\beta) = \begin{bmatrix} \cos\beta & 0 & \sin\beta \\ 0 & 1 & 0 \\ -\sin\beta & 0 & \cos\beta \end{bmatrix}$
        4. $R_z(\gamma) = \begin{bmatrix} \cos\gamma & -\sin\gamma & 0 \\ \sin\gamma & \cos\gamma & 0 \\ 0 & 0 & 1 \end{bmatrix}$
        5. $R = R_z(\alpha)R_y(\beta)R_x(\gamma)$ for arbitrary rotation
    3. Inspection from Learning perspective
        1. Euler Angle is not unique for some rotations. For example $R_z(45^o)R_y(90^o)R_x(45^o) = R_z(90^o)R_y(90^o)R_x(90^o)$
        2. $= \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$
        3. Gimbal lock:
        4. Df is rank-deficient at some $\theta \Rightarrow$ some movement in $T_{f(\theta)}(SO(3))$ cannot be achieved.
        5. For example, when $\beta = \pi/2$, $R = R_z(\alpha)R_y(\pi/2)R_x(\gamma)$
        6. $= \begin{bmatrix} 0 & 0 & 1 \\ \sin(\alpha+\gamma) & \cos(\alpha+\gamma) & 0 \\ -\cos(\alpha+\gamma) & \sin(\alpha+\gamma) & 0 \end{bmatrix}$
        7. since changing α and γ has the same effects, a degree of freedom disappears!

	Inverse?	Composing?	Any local movement in SO(3) can be achieved by local movement in the domain?
Rotation Matrix	✔	✔	N/A
Euler Angle	Complicated	Complicated	No
Angle-axis	✔	Complicated	?
Skew-symmetrical Matrix	✔	Complicated	?
Quaternion	✔	✔	✔

? means no singularity with single exceptions

Related: [Scipy]

References:

1. ML meets geometry, Hao Su Lab, UCSD, 2022.